Growing up in a compound for doctors due to my dad's job, and with friends studying medicine, I developed an interest in healthcare early on. The first time I worked on the healthtech intersection was after studying ML in my bachelor's. My mind went to how biases in healthcare literature would be cemented in the upcoming age of AI healthcare. I compiled my findings into a blog and messaged researchers working on such models at the time, proposing methods to check fairness. Received a few positive responses, but a lack of resources prevented any actual work. That's what motivated me to pursue a master's and be part of an institute again.

Satisfied with my ML knowledge base, I focused on applying it. I took a health innovations course covering the EU healthtech landscape and the end-to-end process of bringing innovations through it. A hospital project course investigating EHR interoperability challenges in Sweden, learning the gap between implemented and existing standards. A small consultancy project forecasting patient arrivals at an emergency department. Most recently, I've started a pilot with a hospital in Saudi Arabia on using AI to read electrophoresis results and speed up lab turnaround times. None are deep research projects individually, but the breadth of exposure means I understand the structures and common problems. A PhD coming from that awareness is more likely to produce something workers actually want to use, and that fits the system.

My master's thesis includes an annotated dataset of short-form video comments I built and a benchmark of classification approaches, including a two-stage multimodal pipeline for image-based comments. It required working directly with label imbalance, annotation disagreement and producing a custom architecture. I'm proud of that last part, because I noticed that it was easier to label comments as harmful or not based on the overall sentiment in the comment section, so why not include multiple passes in the prediction architecture?

The core problem in the PhD is similar to what I faced in my thesis. Labelling comments as harmful, comments that the platforms clearly decided don't break their ToS, was difficult. Cases where nearly identical comments on separate posts have different meanings, and figuring out how to translate this for the model outside of the base encodings. While "harm" is a lot more subjective than the clean datasets you have on AMP actions, the end result is the same. Model architectures have always relied on producing large, clean and relevant data to train on. Semi-supervised learning would normally be used here, and is often used for labelling. I'm curious about how well it can handle complex omics data.

I don't have a background in molecular biology or bioinformatics. I understand that the position is open to strong ML candidates who can acquire the domain knowledge through coursework. I bring experience working with structured noisy data and multi-label classification under label uncertainty. What I'd want to understand more about is how the three curated datasets were constructed and what decisions were made about the label ontology. That's where I'd expect the most interesting design choices to sit.

Lastly, growing microbial resistance is close to me, as Pakistan both causes and suffers casualties from this impending disaster. Any work towards solutions is impactful and very necessary research that needs to be paid more attention to.